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AMRITA SCHOOL OF ENGINEERING, BENGALURU

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

B. TECH. V SEMESTER (ECE AND EAC) AY 2026-2027

ASSIGNMENT FOR 23ECE302 VLSI DESIGN

SYSTEM DESIGN FOR 23ECE383 VLSI DESIGN LABORATORY

REPORT

JULY TO NOVEMBER 2026

Implementation of a Compact Multiphase Divide-by-Two CMOS Frequency Divider in 45 nm CMOS (Cadence GPDK045)

Submitted by

REGISTER NUMBER	*NAME*	*SIGNATURE WITH DATE*

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FACULTY EXAMINER SIGNATURE WITH EVALUATION DATE

TITLE OF THE SYSTEM DESIGN PROJECT:

Implementation of a Compact Multiphase Divide-by-Two CMOS Frequency Divider in 45 nm CMOS (Cadence GPDK045)

TITLE OF THE PAPER:

A Compact Multiphase Divide-by-Two CMOS Frequency Divider

DETAILS OF THE PAPER (SOURCE OF THE TITLE):

Authors: Y.-F. Kuo et al.
Journal: IEEE Microwave and Wireless Technology Letters
Volume: 35
Issue: 5
Pages: 617-620
Date: May 2025
DOI: 10.1109/LMWT.2025.3528863

CIRCUIT DIAGRAM:

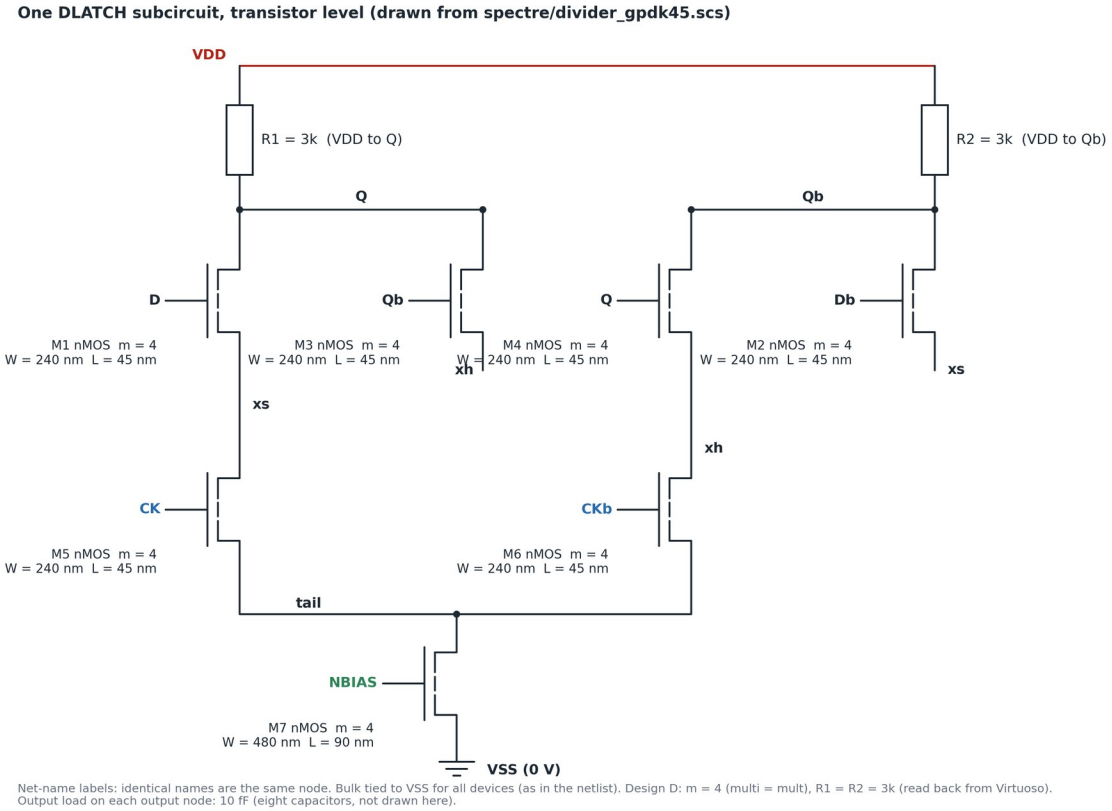


Figure 1: Circuit diagram of one differential DLATCH of design D (M1 to M7, R1, R2), drawn by scripts/draw_gpdk45_circuits.py from spectre/divider_gpdk45.scs. Labels: M1 to M6 W = 240 nm, L = 45 nm; M7 W = 480 nm, L = 90 nm; m = 4; R1 = R2 = 3 kOhm. Generated drawing, not a Cadence screenshot.

THEORY AND WORKING PRINCIPLE:

Topology. The divider follows Kuo et al. (References): four cascaded differential D-latches with alternating cross-connections Q1b to D2, Q2 to D3, Q3b to D4 and Q4 to D1b. Latches 2 and 3 have CK and CKb swapped. Each latch has a sampling pair M1 and M2, a positive-controlled switch M5, a hold pair M3 and M4, a negative-controlled switch M6, a tail M7 and loads R1 and R2. While CK is high a latch samples; while CK is low the hold pair regenerates the stored value. The ring connections make the eight outputs toggle in pairs at half the input frequency, spread around one period.

Elmore estimate. In the paper's model the latch delay is t_pdf = t_pdr = (8C)(R/2) = 4RC with R1 = R/2, so R = 2 R1, and C is the node capacitance; node b is grounded because M7 is on. The paper requires T_CK/2 > 2 Td. With the 10 fF output load, Td = 240 ps for design D against T_CK/2 = 357 ps at 1.4 GHz (bound not met), and Td = 160 ps for design B against T_CK/2 = 83 ps at 6 GHz (bound not met). Both designs divide in simulation, so this bound is not the limit

that decides division here. The Elmore expression is used only for the trend of the sizing, not as a sufficient design rule.

Stack resistance and regeneration. The output pull-down path is a stack of three series devices (M1, M5, M7), so its resistance is about 3 Ron/m. Regeneration of the hold pair needs gm R1 > 1; the margin used here is R1 >= 1.1 x (3 Ron/m). With Ron = 2.3 kOhm (LEDGER.md) the unit stack is 6.90 kOhm, far above the unsized load of 1.15 kOhm, which is why the unsized circuit does not divide. For m = 4 the threshold is 1.90 kOhm (band 1.88 to 2.25 kOhm), so design D at 3 kOhm passes, while design B at 2 kOhm lies inside the band and its margin depends on the Ron value used. The recorded division results in LEDGER.md agree with this rule for 5 of 5 numeric (m, R) pairs.

Phase-step law. Outputs come in pairs. The steps between neighbouring outputs alternate between a and 90 - a degrees, with a = 360 t_pair f_in. At 1.4 GHz the measured steps are 83.46 and 6.54 deg (a = 6.54 deg, t_pair = 13.0 ps); at 6 GHz the steps are 48.78 and 41.22 deg (a = 41.22 deg, t_pair = 19.1 ps). Equal 45 deg steps need T_CK = 8 t_pair. Because t_pair differs between the two designs, the phase error is a property of each design at a given f_in; in the frequency sweeps it falls as f_in rises and is small only near each design's frequency limit. The paper's 3.75 deg is reproduced in magnitude only by design B at 6 GHz.

Delay and power definitions. Delay is the mean over CK rising edges of the time to the next rising edge on any of the eight outputs. Power is VDD times the mean supply current over whole output periods. PDP is power times delay (mW times ps gives fJ). A division is accepted only when all eight outputs run at f_in/2 within 0.1 percent and each swings at least 0.1 V.

TOOL & TECHNOLOGY USED:

Simulator: Cadence Spectre 12.1.0 (MMSIM121), run on a 32-bit CentOS 6.5 virtual machine. Layout and DRC tools in the flow: Virtuoso IC615 and Assura 4.1.

Process: Cadence GPDK045 v6.0, a 45 nm generic PDK. Device g45n1svt (Virtuoso cell nmos1v), a 1.0 V thin-oxide standard-Vt NMOS modelled with BSIM4. Corners tt, ff, ss, fs and sf from models/spectre/gpdk045.scs.

Sizes: unit transistor W = 240 nm, L = 45 nm for M1 to M6; M7 W = 480 nm, L = 90 nm; the multiplier m scales every device in parallel.

Stimulus and conditions: VDD = 1.0 V with PVT at 0.9 and 1.1 V; temperature 27 C nominal; CK rise and fall 20 ps; CKb is CK delayed by half a period; tail bias vb = 1 V; 10 fF load on each of the eight outputs.

Virtuoso database read-back of the DLATCH of design D (results/gpdk045/virtuoso_dlatch_params_designD.txt): R1 and R2 = 3 kOhm; M1 to M6 w = 240n, l = 45n, m = 4; M7 w = 480n, l = 90n, m = 4. These agree with the Spectre netlist sizes and with Tables 1 and 2.

Plots are generated by scripts from the Spectre results; the waveform and PVT figures are plots, not Virtuoso screenshots, and each caption says so. The two circuit drawings (Figure 1 and Figure 9) are generated by scripts/draw_gpdk45_circuits.py from spectre/divider_gpdk45.scs.

SCREENSHOT OF THE SIMULATOR BASED CIRCUIT IMPLEMENTATION:

Unit devices (m = 1) with the paper's rule applied literally: R1 = R2 = Ron/2 = 1.15 kOhm, where Ron = 2.3 kOhm is the measured on-resistance of the unit nMOS (LEDGER.md; the Vds sweep gives 2.28 to 2.73 kOhm). Device sizes are read from spectre/divider_gpdk45.scs. At 1.4 GHz the eight outputs follow the clock: the period-measured Q1 frequency is 2799.8 MHz (about twice f_in) but the Q1 swing is only 15.3 mV, so these are not toggling logic outputs and the circuit does not divide. Delay, PDP and phase error are therefore n/a, and the power of a non-dividing circuit is not reported as a result.

Table 1: Device list, circuit without the sizing concept

Device	**W**	**L**	**Multiplier**	**Role**
M1, M2	240 nm	45 nm	1	sampling pair (differential inputs)
M3, M4	240 nm	45 nm	1	hold pair (cross-coupled regenerative pair)
M5	240 nm	45 nm	1	switch controlled by CK
M6	240 nm	45 nm	1	switch controlled by CKb
M7	480 nm	90 nm	1	tail current source, gate at vb = 1 V
R1, R2	n/a	n/a	n/a	load resistors, 1.15 kOhm each

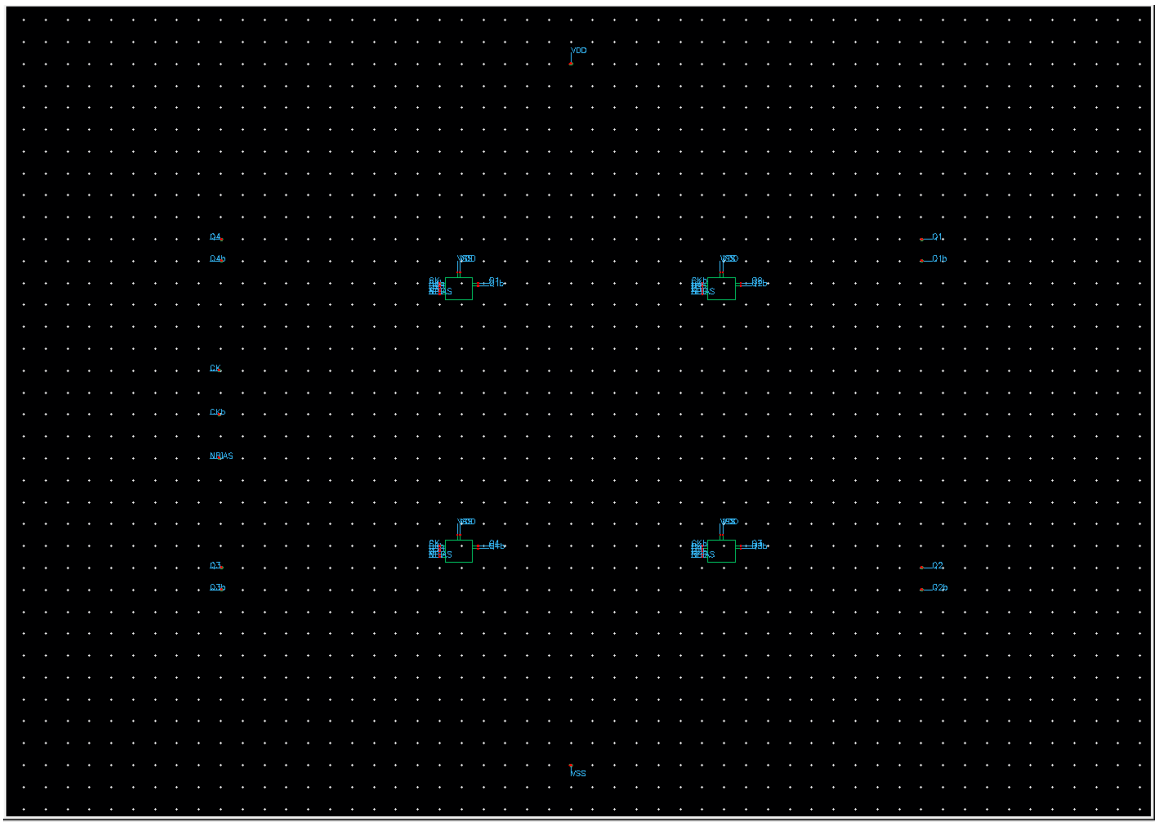


Figure 2: Virtuoso schematic of the divider core (design D library): four DLATCH instances XD1 to XD4 with the ring connections (Cadence GUI screenshot). Device sizes are in Tables 1 and 2.

SCREENSHOT OF THE OUTPUT WAVEFORM OBTAINED FROM THE SIMULATOR (DC AND TRANSIENT):

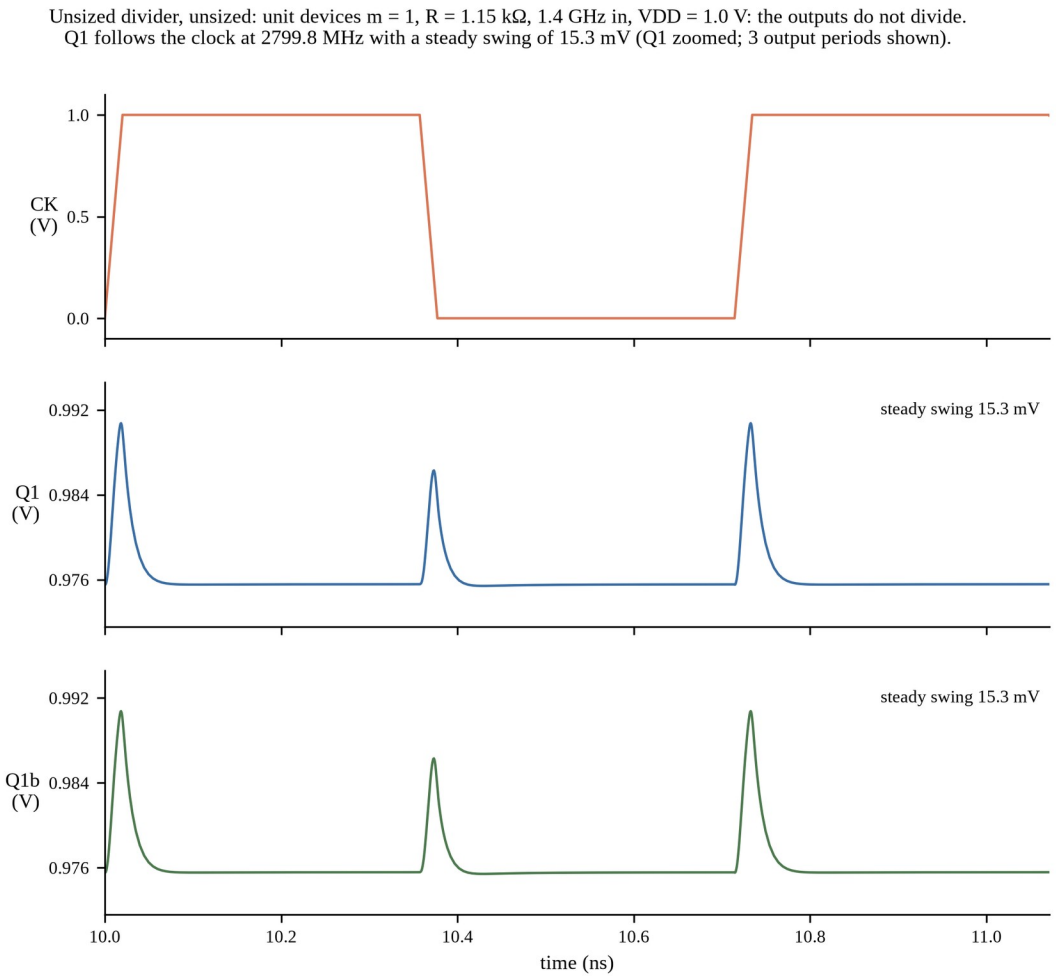


Figure 3: Transient waveforms of the circuit without sizing (m = 1, R1 = R2 = 1.15 kOhm) at 1.4 GHz, plotted from Spectre output (not a simulator screenshot). Q1 swings between 0.975 V and 0.991 V (15.3 mV), so the outputs do not divide. DC operating point (analysis, t = 0, CK low): M7 region triode, Vds = 56.3 nV, Ids = 35.6 pA, Vgs = 1 V.

SCREENSHOT OF THE CIRCUIT BASED ON SIZING CONCEPT:

Same topology with the sizing concept: multiplier m = 4 scales every device in parallel, load R1 = R2 = 3 kOhm, input 1.4 GHz (design D). Stack rule (section Theory): the pull-down stack resistance is 3 Ron/m = 1.72 kOhm, the regeneration threshold 1.1 times that is 1.90 kOhm (band 1.88 to 2.25 kOhm for the Ron range), and the load 3 kOhm is above the band. The sizing check at 1.4 GHz and the 6 GHz point (design B, m = 4, R1 = R2 = 2 kOhm, reported in the other information section) use the same rule. The tail device M7 is in the triode region in the DC analysis (Vds stays below Vdsat), see the other information section.

Table 2: Device list, circuit based on the sizing concept

Device	W	L	Multiplier	Role
M1, M2	240 nm	45 nm	4	sampling pair (differential inputs)
M3, M4	240 nm	45 nm	4	hold pair (cross-coupled regenerative pair)
M5	240 nm	45 nm	4	switch controlled by CK
M6	240 nm	45 nm	4	switch controlled by CKb
M7	480 nm	90 nm	4	tail current source, gate at vb = 1 V
R1, R2	n/a	n/a	n/a	load resistors, 3 kOhm each

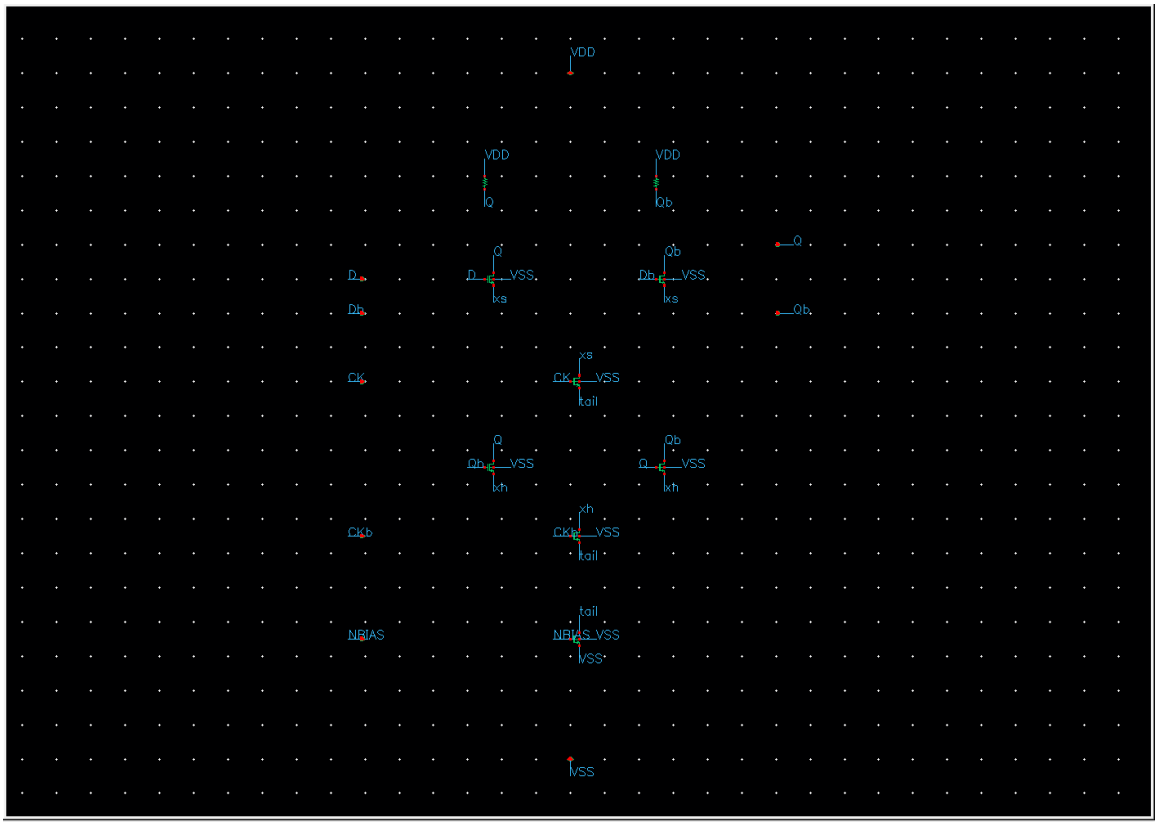


Figure 4: Virtuoso schematic of the DLATCH of design D with sizing: m = 4, R1 = R2 = 3 kOhm (Cadence GUI screenshot, cropped). Device parameters read back from the Virtuoso database are given in the Tool and Technology section and agree with Table 2.

SCREENSHOT OF THE OUTPUT WAVEFORM OBTAINED FROM THE SIMULATOR FOR THE CIRCUIT WITH SIZING CONCEPT (DC AND TRANSIENT):

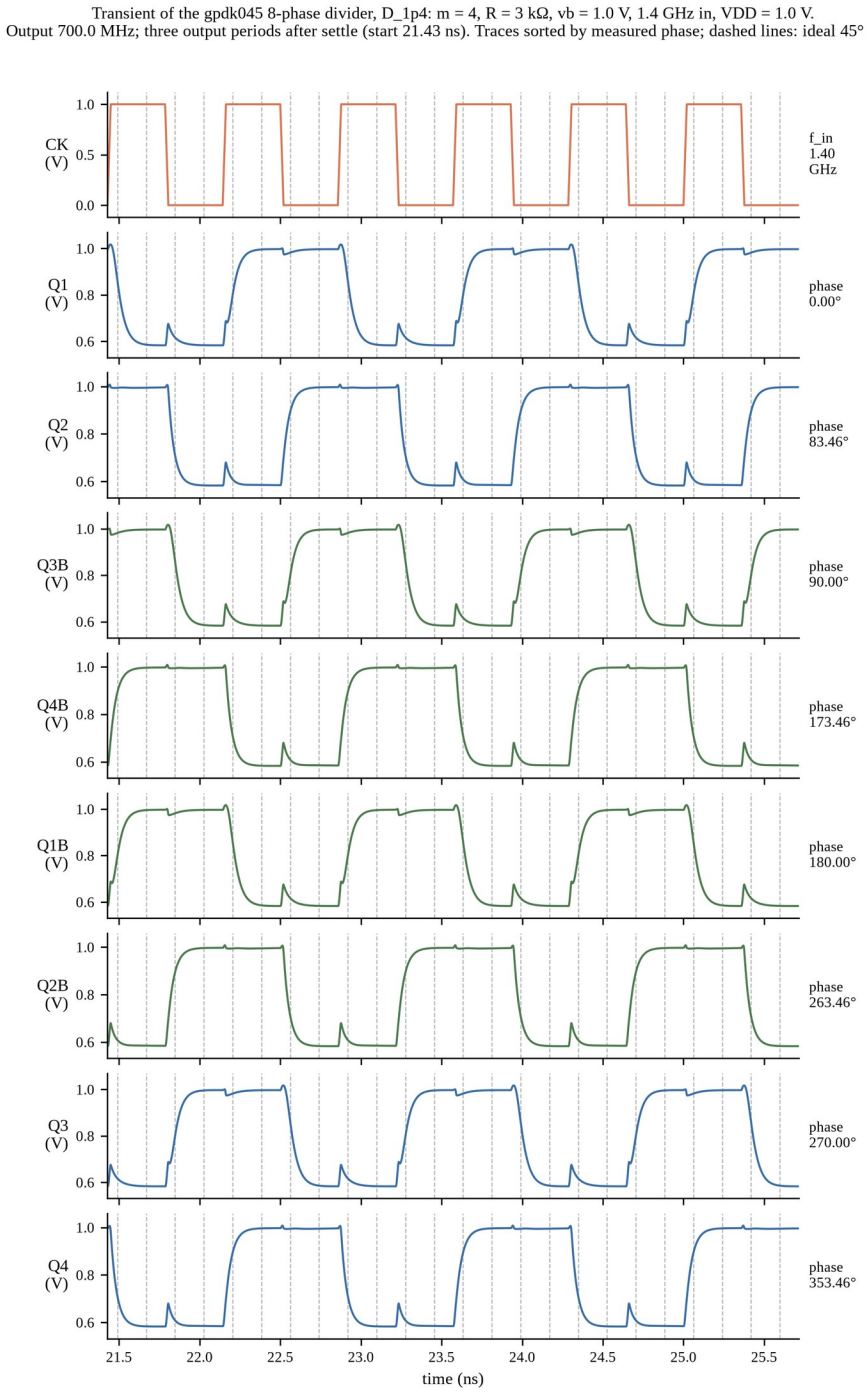


Figure 5: Transient waveforms of design D (m = 4, R1 = R2 = 3 kOhm) at 1.4 GHz, plotted from Spectre output (not a simulator screenshot). Q1 swings between 0.583 V and 1.017 V; output 700.0 MHz; tail voltage stays between 8.1 mV and 66.9 mV.

PROVIDE THE VALUES FOR THE FOLLOWING ALONG WITH THE SCREENSHOTS AS PROOF:

Table 3: Performance values with and without the sizing concept, with the proof figures for each row

Parameter	Value without Sizing	Value with Sizing	Proof (figure)
Power calculation (P)	n/a	0.5413 mW	Fig. 3 (without), Fig. 5 (with)
Delay calculation (t_pd)	n/a	25.99 ps	Fig. 3 (without), Fig. 5 (with)
Power Delay Product (PDP)	n/a	14.067 fJ	Fig. 3 (without), Fig. 5 (with)
Input frequency (f_in)	1.4 GHz	1.4 GHz	Fig. 3 (without), Fig. 5 (with)
Output frequency (f_out)	n/a	700.0 MHz	Fig. 3 (without), Fig. 5 (with)
Max phase-step error	n/a	38.46 deg	Fig. 3 (without), Fig. 5 (with)
Output swing	0.0153 V	0.4344 V	Fig. 3 (without), Fig. 5 (with)
Tail-device saturation check (M7)	n/a	**Vds = 0.0669 V, Vdsat = 0.3578 V: NOT saturated**	Fig. 3 (without), Fig. 5 (with)

SCREENSHOT OF THE PVT ANALYSIS:

Table 4: PVT analysis results (FAIL rows are kept)

Corner	Temp (C)	VDD (V)	Divides	f_out (MHz)	Delay (ps)	Power (mW)	PDP (fJ)	Note
D tt	-40	0.9	PASS	700.0	24.61	0.4356	10.72	
D tt	-40	1.0	PASS	700.0	24.01	0.6892	16.54	
D tt	-40	1.1	PASS	700.0	24.19	0.9714	23.5	
D tt	27	0.9	PASS	700.0	27.7	0.3387	9.38	
D tt	27	1.0	PASS	700.0	25.99	0.5413	14.07	
D tt	27	1.1	PASS	700.0	25.38	0.7769	19.72	
D tt	125	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5001 not 2
D tt	125	1.0	PASS	700.0	29.81	0.3898	11.62	

Corner	Temp (C)	VDD (V)	Divides	f_out (MHz)	Delay (ps)	Power (mW)	PDP (fJ)	Note
D tt	125	1.1	PASS	700.0	28.3	0.5648	15.98	
D ff	-40	0.9	PASS	700.0	24.65	0.5512	13.59	
D ff	-40	1.0	PASS	700.0	24.66	0.8104	19.99	
D ff	-40	1.1	PASS	700.0	25.2	1.0784	27.17	
D ff	27	0.9	PASS	700.0	26.94	0.4303	11.59	
D ff	27	1.0	PASS	700.0	25.97	0.6467	16.8	
D ff	27	1.1	PASS	700.0	25.81	0.891	23.0	
D ff	125	0.9	PASS	700.0	31.05	0.3074	9.55	
D ff	125	1.0	PASS	700.0	29.11	0.4677	13.61	
D ff	125	1.1	PASS	700.0	28.04	0.6547	18.36	
D ss	-40	0.9	PASS	700.0	25.6	0.3147	8.06	
D ss	-40	1.0	PASS	700.0	23.84	0.5483	13.07	
D ss	-40	1.1	PASS	700.0	23.54	0.8217	19.34	
D ss	27	0.9	PASS	700.0	29.71	0.2463	7.32	
D ss	27	1.0	PASS	700.0	26.54	0.4288	11.38	
D ss	27	1.1	PASS	700.0	25.27	0.6476	16.37	
D ss	125	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5001 not 2
D ss	125	1.0	PASS	700.0	30.93	0.3095	9.57	
D ss	125	1.1	PASS	700.0	28.94	0.4688	13.57	
D fs	-40	0.9	PASS	700.0	24.25	0.5072	12.3	
D fs	-40	1.0	PASS	700.0	24.08	0.7669	18.46	
D fs	-40	1.1	PASS	700.0	24.52	1.0436	25.59	
D fs	27	0.9	PASS	700.0	26.74	0.3971	10.62	
D fs	27	1.0	PASS	700.0	25.59	0.6098	15.61	
D fs	27	1.1	PASS	700.0	25.29	0.8525	21.56	
D fs	125	0.9	**FAIL**	n/a	n/a	n/a	n/a	q1 swing 0.093 V < 0.1 V
D fs	125	1.0	PASS	700.0	28.89	0.4423	12.78	
D fs	125	1.1	PASS	700.0	27.72	0.6259	17.35	
D sf	-40	0.9	PASS	700.0	25.38	0.3632	9.22	
D sf	-40	1.0	PASS	700.0	24.16	0.6055	14.63	
D sf	-40	1.1	PASS	700.0	24.03	0.8837	21.23	
D sf	27	0.9	PASS	700.0	29.2	0.2811	8.21	
D sf	27	1.0	PASS	700.0	26.63	0.4713	12.55	
D sf	27	1.1	PASS	700.0	25.65	0.6964	17.86	
D sf	125	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5001 not 2
D sf	125	1.0	PASS	700.0	30.89	0.3375	10.43	
D sf	125	1.1	PASS	700.0	29.09	0.5023	14.61	
B6 tt	-40	0.9	PASS	3000.0	19.09	0.3881	7.41	
B6 tt	-40	1.0	PASS	3000.0	16.42	0.6424	10.55	

Corner	Temp (C)	VDD (V)	Divides	f_out (MHz)	Delay (ps)	Power (mW)	PDP (fJ)	Note
B6 tt	-40	1.1	PASS	3000.0	15.79	0.9453	14.92	
B6 tt	27	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 tt	27	1.0	PASS	3000.0	27.94	0.4929	13.77	
B6 tt	27	1.1	PASS	3000.0	19.15	0.7327	14.03	
B6 tt	125	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 tt	125	1.0	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 tt	125	1.1	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 ff	-40	0.9	PASS	3000.0	17.36	0.5147	8.93	
B6 ff	-40	1.0	PASS	3000.0	16.28	0.7959	12.96	
B6 ff	-40	1.1	PASS	3000.0	16.06	1.1231	18.04	
B6 ff	27	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 ff	27	1.0	PASS	3000.0	20.52	0.6124	12.57	
B6 ff	27	1.1	PASS	3000.0	18.42	0.8744	16.1	
B6 ff	125	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 ff	125	1.0	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 ff	125	1.1	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 ss	-40	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 ss	-40	1.0	PASS	3000.0	17.46	0.4902	8.56	
B6 ss	-40	1.1	PASS	3000.0	15.87	0.7623	12.1	
B6 ss	27	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 ss	27	1.0	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 ss	27	1.1	PASS	3000.0	21.78	0.5899	12.85	
B6 ss	125	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 ss	125	1.0	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 ss	125	1.1	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 fs	-40	0.9	PASS	3000.0	17.48	0.4638	8.11	
B6 fs	-40	1.0	PASS	3000.0	16.04	0.735	11.79	
B6 fs	-40	1.1	PASS	3000.0	15.73	1.0534	16.57	
B6 fs	27	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000

Corner	Temp (C)	VDD (V)	Divides	f_out (MHz)	Delay (ps)	Power (mW)	PDP (fJ)	Note
								not 2
B6 fs	27	1.0	PASS	3000.0	20.89	0.5681	11.87	
B6 fs	27	1.1	PASS	3000.0	18.3	0.8222	15.04	
B6 fs	125	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 fs	125	1.0	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 fs	125	1.1	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 sf	-40	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.9967 not 2
B6 sf	-40	1.0	PASS	3000.0	17.17	0.5509	9.46	
B6 sf	-40	1.1	PASS	3000.0	15.99	0.8357	13.37	
B6 sf	27	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 sf	27	1.0	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 sf	27	1.1	PASS	3000.0	20.79	0.6435	13.38	
B6 sf	125	0.9	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 sf	125	1.0	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2
B6 sf	125	1.1	**FAIL**	n/a	n/a	n/a	n/a	ratio 0.5000 not 2

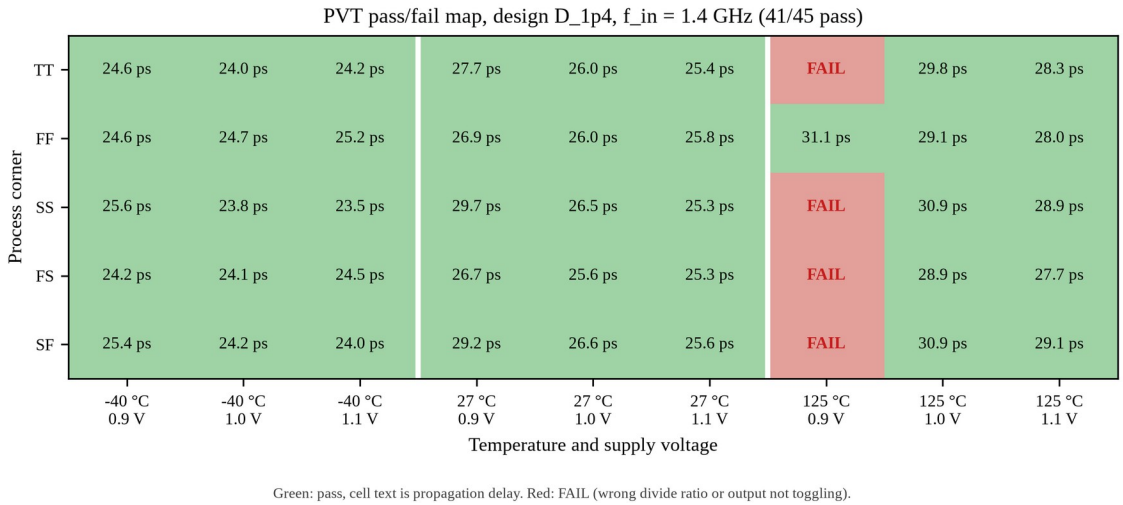


Figure 6: PVT pass/fail map of design D at 1.4 GHz (5 corners x -40/27/125 C x 0.9/1.0/1.1 V), plotted from pvt_D.tsv, not a Cadence screenshot. 41 of 45 points pass.

SCREENSHOT OF THE LAYOUT:

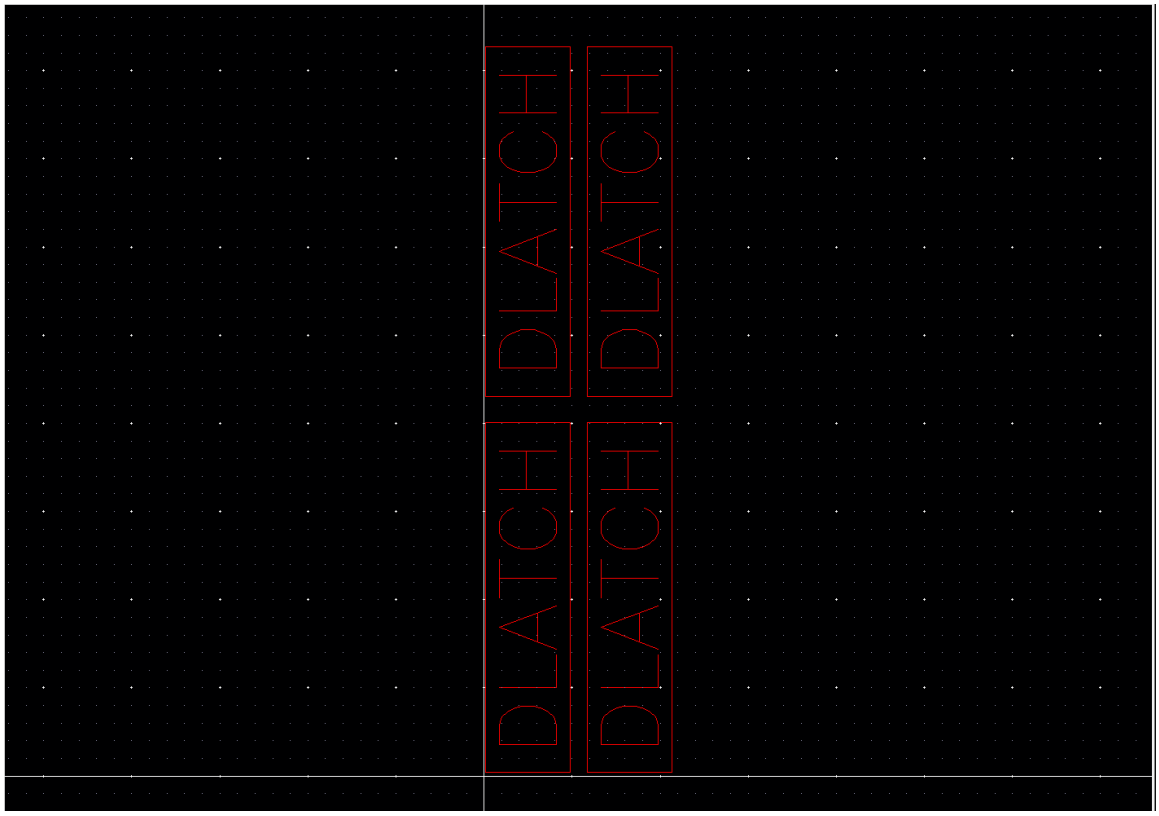


Figure 7: Virtuoso layout of divider_core: four DLATCH instances (XD1 to XD4) placed in a 2x2 arrangement, placement only (Virtuoso Layout Suite screenshot, cropped to the layout canvas). Routing is not drawn.

SCREENSHOT OF THE OUTPUT WAVEFORM OBTAINED FROM THE SIMULATOR FOR THE LAYOUT:

NOT YET AVAILABLE: Figure 8 - simulator waveform screenshot of the layout

ANY OTHER RELEVANT INFORMATION:

Layout: what was done (placement only). The layout was generated by the SKILL script cadence/skill/build_layout_v2.il in Virtuoso 6.1.5 (IC615). The DLATCH uses the PDK pcells nmos1v (two-finger, fw = 120 nm per unit; M7 has four fingers) and resnsnpoly for the two poly resistors R1 and R2 (about 2 kOhm at 200 ohm/sq, segW 0.8 um, segL 7.855 um). This resistor size is a placeholder: the simulated design D uses 3 kOhm, so the layout resistor is not the simulated value. The DLATCH box is 4.875 x 20.28 um. divider_core places XD1 R0, XD2 MY, XD3 MX and XD4 R180 with a 1 um gap, and the clock-star lengths are equal (98.62 um each, difference 0). Figure 7 shows the divider_core layout and Figure 21 the DLATCH layout.

Layout: Assura DRC (real result). Assura 4.1 DRC was run on the DLATCH layout only (divider_lib/DLATCH/layout, gpdk045 assuraDRC.rul); divider_core was not run. The result is 21 errors in 3 rules (Figure 22). Rule 220 psubstrate_StampErrorMult (1) and Rule 221 psubstrate_StampErrorConnect (6) give 7 substrate-stamp connectivity errors, because no tap or substrate connection is routed. Rule 595 METAL1.A.1, Metal1 area must be at least 0.02 um (14), gives 14 Metal1 minimum-area errors on the pin shapes. The DLATCH layout is therefore not DRC clean.

Layout: what was not done. The layout is placement only. Intra-latch metal routing, vias, clock trunks and substrate taps or connections are not drawn. LVS, parasitic extraction and post-layout simulation were not run, so the simulated results in this report are pre-layout results and the layout is not complete.

Virtuoso connectivity check (results/gpdk045/core_connectivity_virtuoso.txt): the top-level connections of 4 DLATCH instances (XD1 to XD4) were compared pin by pin with spectre/divider_gpdk45.scs, giving 0 mismatches. Schematic checks (schCheck errors/warnings, LEDGER.md): DLATCH 0/0, divider_core 0/0, tb_divider 0/1.

Second operating point, design B at 6 GHz: m = 4, R1 = R2 = 2 kOhm, output 3000.0 MHz, power 0.4929 mW, delay 27.94 ps, PDP 13.768 fJ, maximum phase-step error 3.78 deg (steps 48.78 and 41.22 deg). Q1 swing 0.818 to 0.953 V. The M7 tail voltage reaches 66.2 mV at most. Vdsat of M7 was not extracted for this point, so its saturation check is n/a. The metrics of this point are given here, not in Table 3; its PVT matrix is listed in Table 4 under the rows labelled B6.

DC operating point (analysis at t = 0, CK low, no tail current). Unsized: M7 triode, Vds = 56.3 nV, Ids = 35.6 pA. Design D: M7 triode, Vds = 54.4 nV, Ids = 137.4 pA. Design B: M7 triode, Vds = 54.4 nV, Ids = 137.4 pA (the same as design D, because both designs have m = 4 and vb = 1 V and no tail current flows at t = 0). Vgs = 1 V in all three.

Tail device M7 (honest finding). In the transient, the tail voltage of design D stays between 8.1 mV and 66.9 mV, while Vdsat = 357.8 mV (Vth = 527.5 mV, gm = 200.9 pS, gds = 2.531 mS at the bias). Vds never reaches Vdsat, so M7 operates in the triode region. The paper's tail is a saturated current source; this circuit's tail is not, and the report states this deviation openly (see Figure 12).

Frequency sweeps (fs_B, fs_D, fs_E). Design B divides from 1.4 to 6 GHz and fails at 6.2 GHz, where the output ratio is not 2. Design D (m = 4, R = 3 kOhm) divides from 3 to 5.5 GHz and fails at 6 GHz. Design E (m = 6, R = 2 kOhm) divides from 3 to 7 GHz and fails at 8 GHz. The maximum phase-step error at the top of each dividing range is 3.78 deg (B), 13.82 deg (D) and 15.24 deg (E); at the bottom of the range it is 38.68, 30.89 and 34.32 deg. The error falls with input frequency in all three sweeps (Figure 14).

PVT results. Design D at 1.4 GHz: 41 of 45 pass; 4 fail (125 C: 4): tt_125_0.9, ss_125_0.9, fs_125_0.9, sf_125_0.9. Design B at 1.4 GHz: 27 of 45 pass; 18 fail (27 C: 3, 125 C: 15):

tt_27_0.9, tt_125_0.9, tt_125_1.0, tt_125_1.1, ff_125_0.9, ff_125_1.0, ff_125_1.1, ss_27_0.9, ss_125_0.9, ss_125_1.0, ss_125_1.1, fs_125_0.9, fs_125_1.0, fs_125_1.1, sf_27_0.9, sf_125_0.9, sf_125_1.0, sf_125_1.1. Design E (m = 6, R = 2 kOhm) at 1.4 GHz: 41 of 45 pass; 4 fail (125 C: 4): tt_125_0.9, ss_125_0.9, fs_125_0.9, sf_125_0.9. Design B at 6 GHz: 21 of 45 pass; 24 fail (-40 C: 2, 27 C: 7, 125 C: 15): tt_27_0.9, tt_125_0.9, tt_125_1.0, tt_125_1.1, ff_27_0.9, ff_125_0.9, ff_125_1.0, ff_125_1.1, ss_-40_0.9, ss_27_0.9, ss_27_1.0, ss_125_0.9, ss_125_1.0, ss_125_1.1, fs_27_0.9, fs_125_0.9, fs_125_1.0, fs_125_1.1, sf_-40_0.9, sf_27_0.9, sf_27_1.0, sf_125_0.9, sf_125_1.0, sf_125_1.1. The passing range for design D is power 0.246 to 1.078 mW and delay 23.54 to 31.05 ps. The PVT tables list design D (1.4 GHz) first and design B (6 GHz) second, with the design in the corner column.

Delay cross-check. The single-edge CK-to-Q1 delay in analysis.json is 51.95 ps for design D and 66.10 ps for design B. That is a different quantity (first edge of one output), so it is a cross-check only. The table delay is the mean over all eight outputs, which agrees with analysis.json to within 0.01 ps (25.99 ps for D). Q1 rise and fall times (10 to 90 percent) for D are 134.6 ps and 102.6 ps.

Power cross-check. The mean supply current of design D from analysis.json is 0.54144 mA, against 0.541337 mA in the TSV, within 0.018%. The peak supply current in steady state is 0.560 mA.

Comparison with the paper. Literature values from Kuo et al. (not re-measured here): TSMC 180 nm, 1.8 V, 1.4 GHz input, 2.34 mW, phase error within 3.75 deg, core 91.7 x 52.9 um^2. In this generic 45 nm PDK, design D at 1.4 GHz draws 0.5413 mW but its phase-step error is 38.46 deg; design B reaches 3.78 deg only at 6 GHz. The core area is not measured here and the processes differ, so this report makes no claim of beating the paper.

Figure numbering. Figures 1 to 8 are the named slots. Figures 9 onwards are additional figures: the ring diagram (Figure 9), the 6 GHz waveforms (Figure 10), phase steps (Figure 11), tail voltage (Figure 12), supply current (Figure 13), the frequency sweep (Figure 14) and the PVT plots for design D and B (Figures 15, 16 and 17), the testbench schematic (Figure 18), the Virtuoso session log (Figure 19), the Library Manager (Figure 20), the DLATCH layout (Figure 21) and the Assura DRC tally (Figure 22).

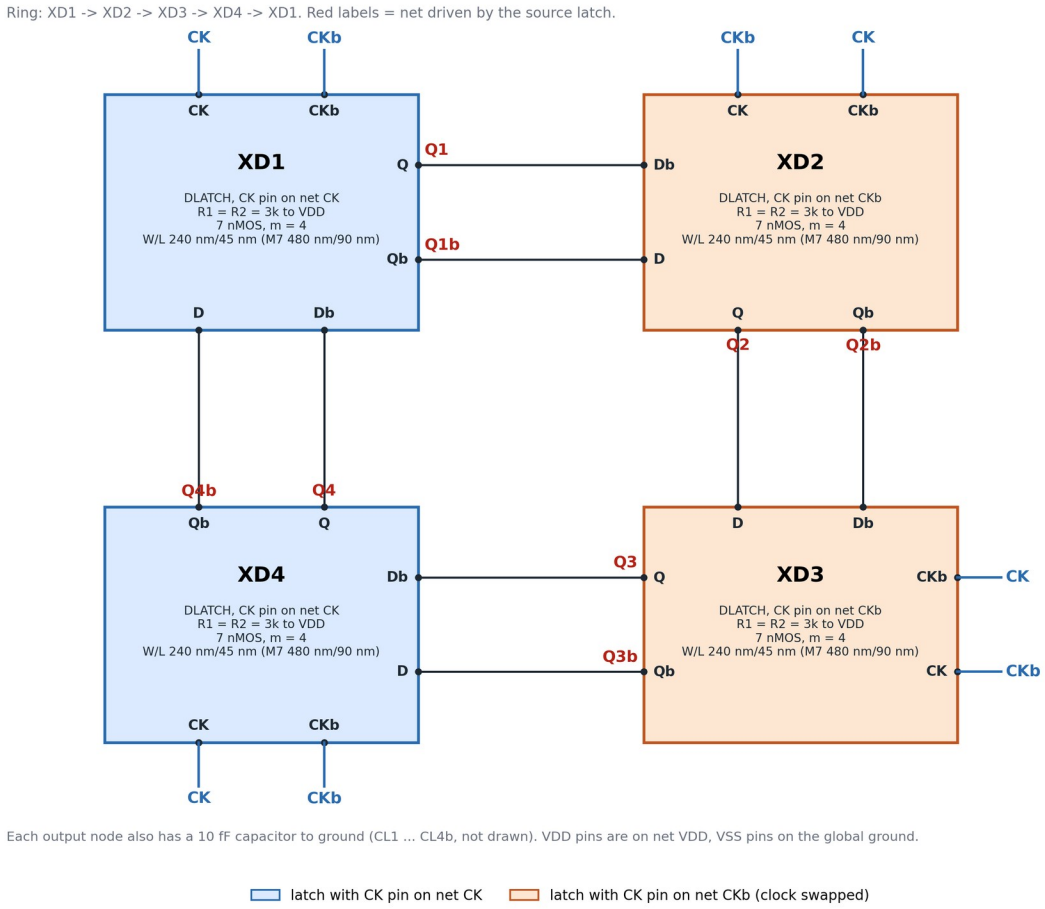


Figure 9: Ring interconnection of the four DLATCH instances XD1 to XD4 (alternating cross-connections, CK and CKb swapped on XD2 and XD3), drawn by scripts/draw_gpdk45_circuits.py from spectre/divider_gpdk45.scs with design D labels (m = 4, R = 3 kOhm). Generated drawing, not a Cadence screenshot.

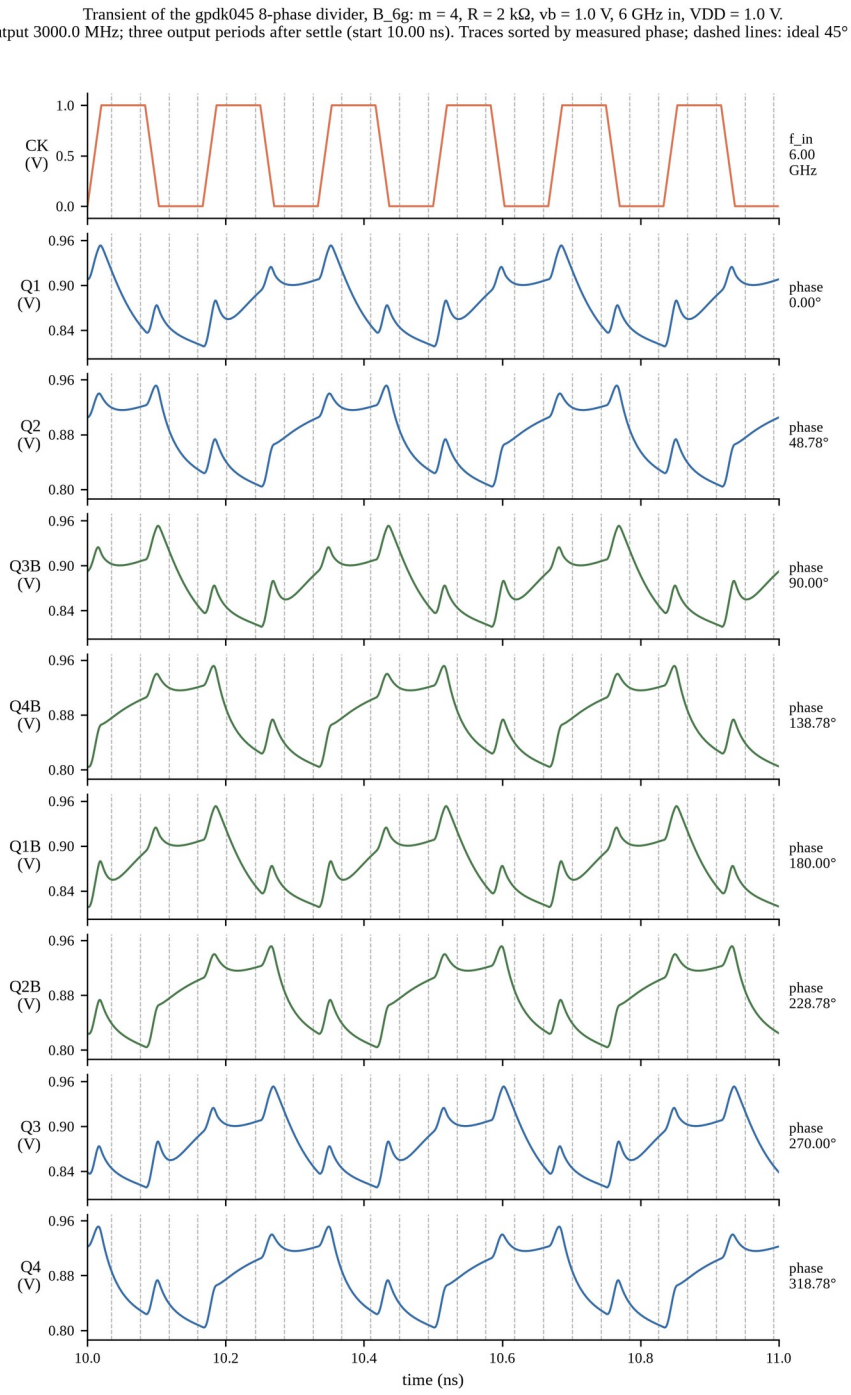


Figure 10: Transient waveforms of design B at 6 GHz (m = 4, R1 = R2 = 2 kOhm), plotted from Spectre output. Output period-measured frequency 3000.0 MHz.

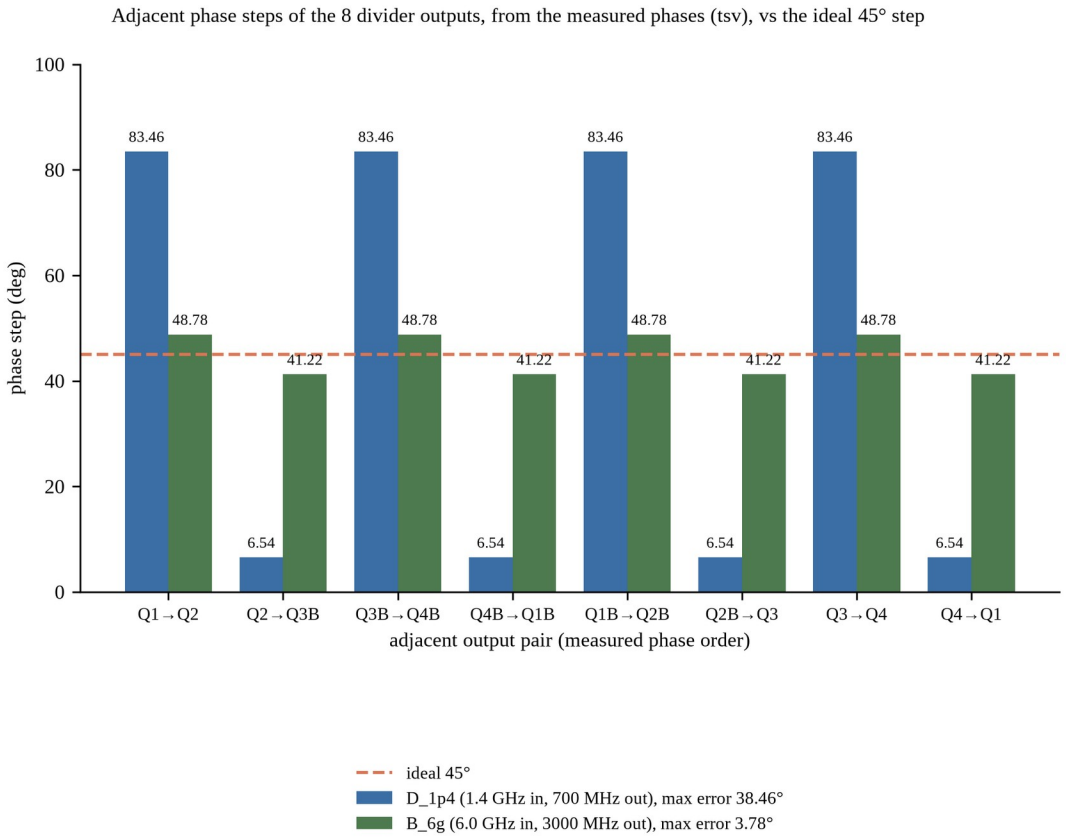


Figure 11: Phase steps between the eight outputs for designs D (1.4 GHz) and B (6 GHz), plotted from the nominal Spectre results.

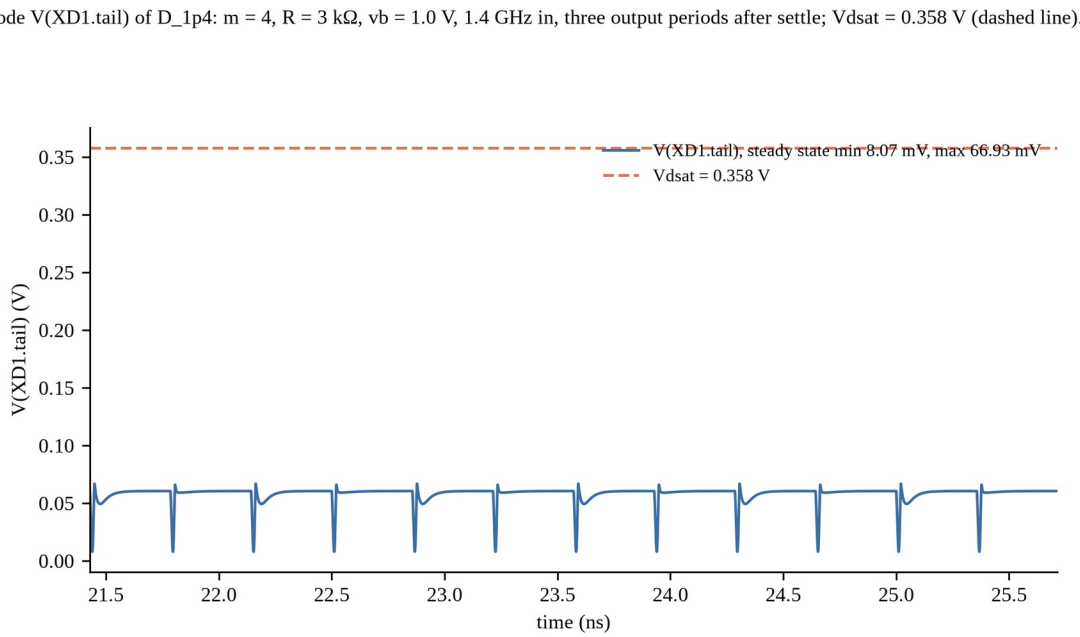


Figure 12: Tail (M7) drain voltage of design D at 1.4 GHz compared with the M7 saturation voltage Vdsat = 0.358 V, plotted from Spectre output.

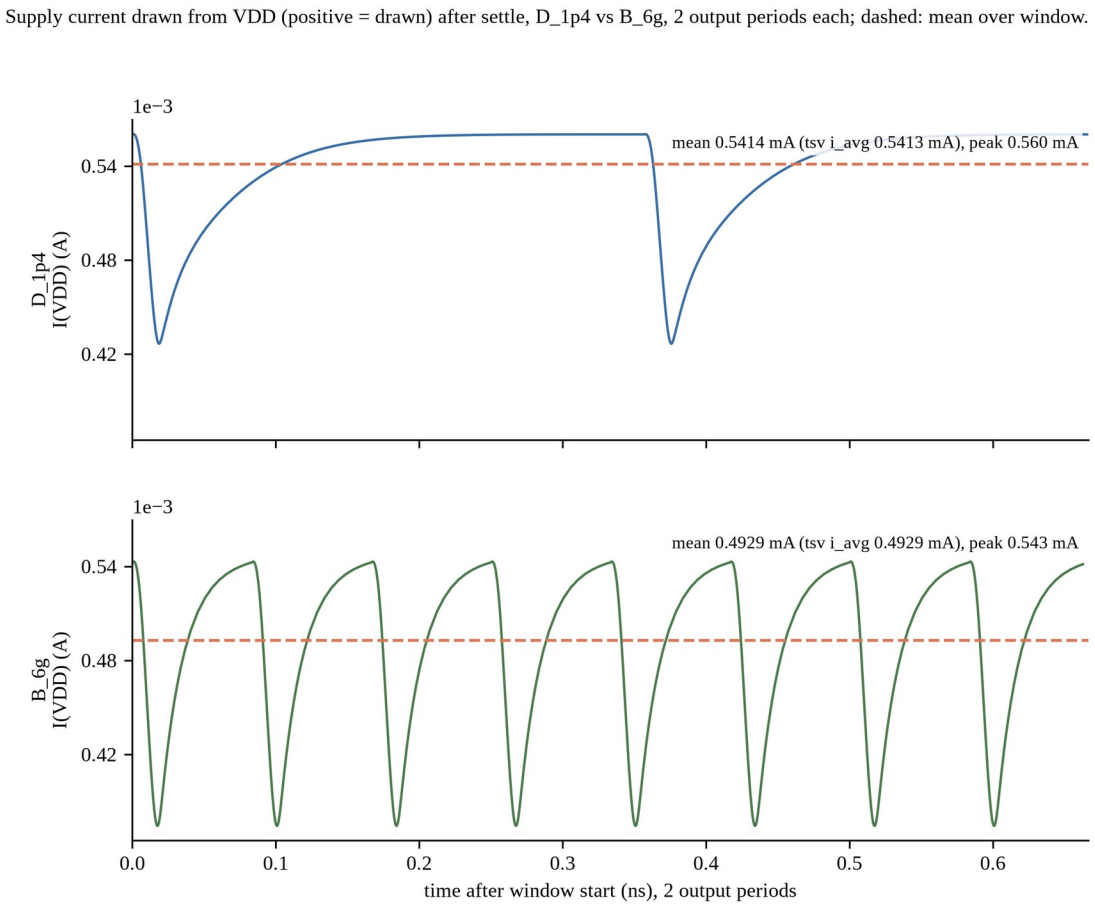


Figure 13: Supply current drawn from VDD, plotted from Spectre output; the table power is VDD times the mean over whole output periods.

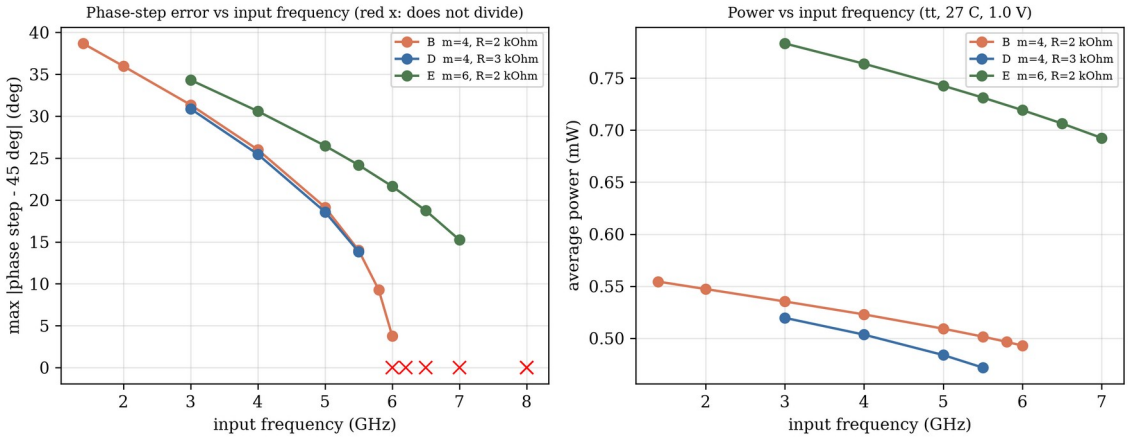


Figure 14: Frequency sweep of designs B, D and E (fs_B.tsv, fs_D.tsv, fs_E.tsv): divided output and maximum phase-step error against input frequency.

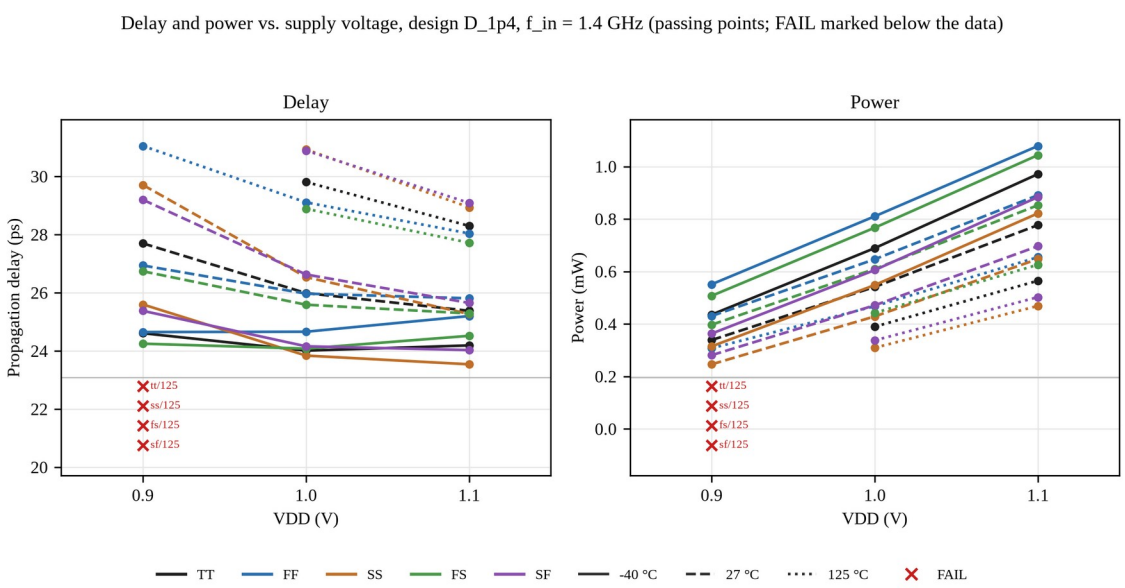


Figure 15: Delay and power over the PVT matrix of design D at 1.4 GHz, plotted from pvt_D.tsv (passing points only).

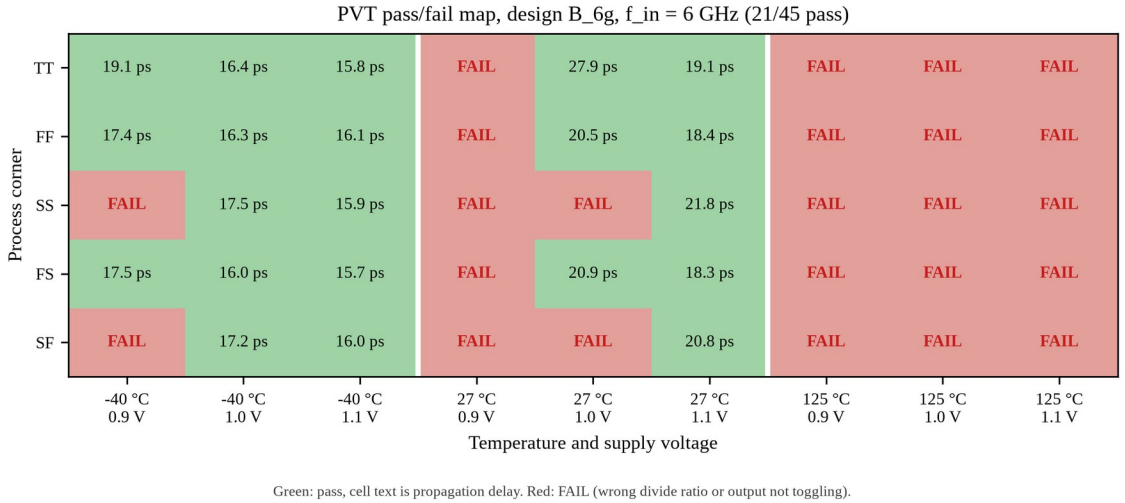


Figure 16: PVT pass/fail map of design B at 6 GHz, plotted from pvt_B6.tsv. 21 of 45 points pass.

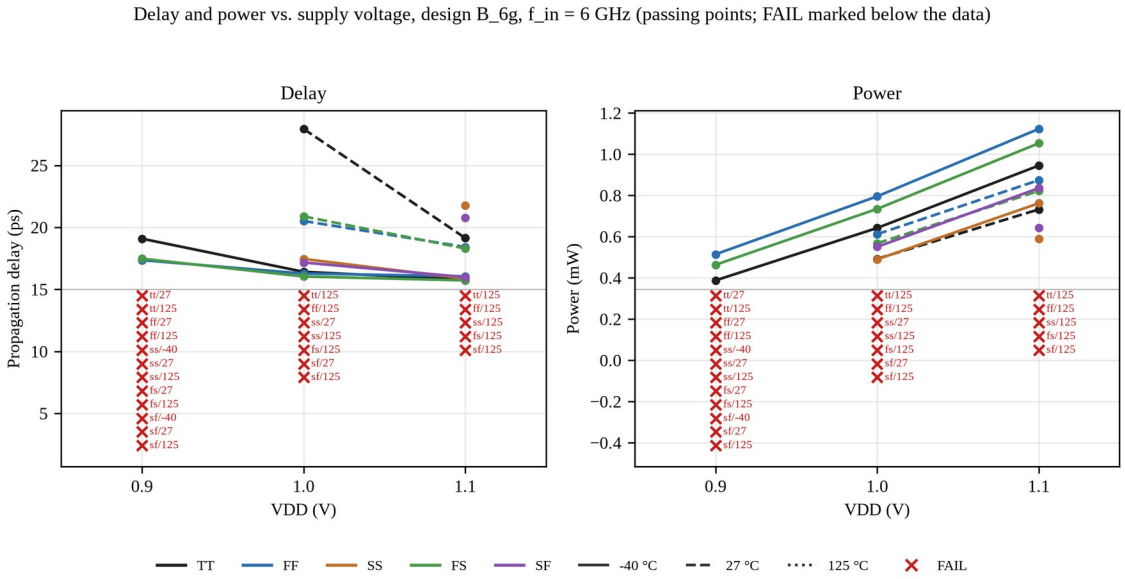


Figure 17: Delay and power over the PVT matrix of design B at 6 GHz, plotted from pvt_B6.tsv (passing points only).

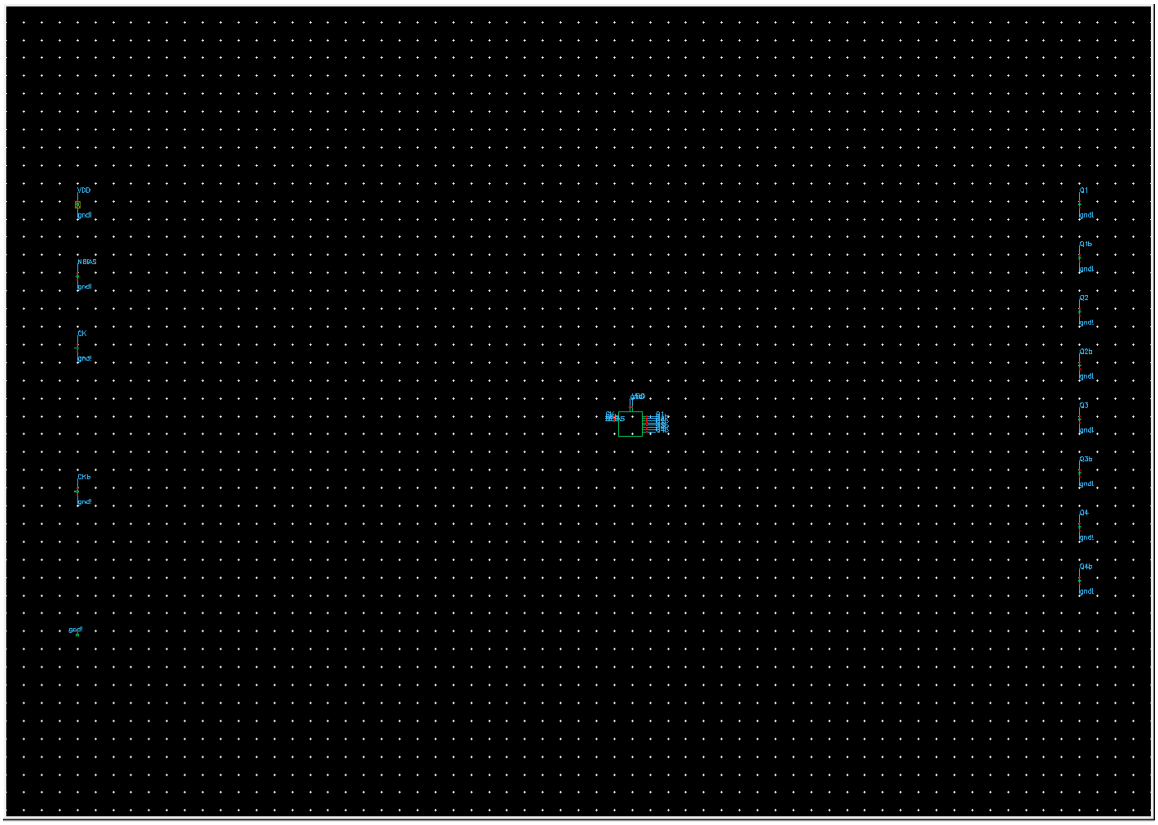


Figure 18: Virtuoso schematic of the testbench tb_divider (Cadence GUI screenshot, cropped to the schematic window).

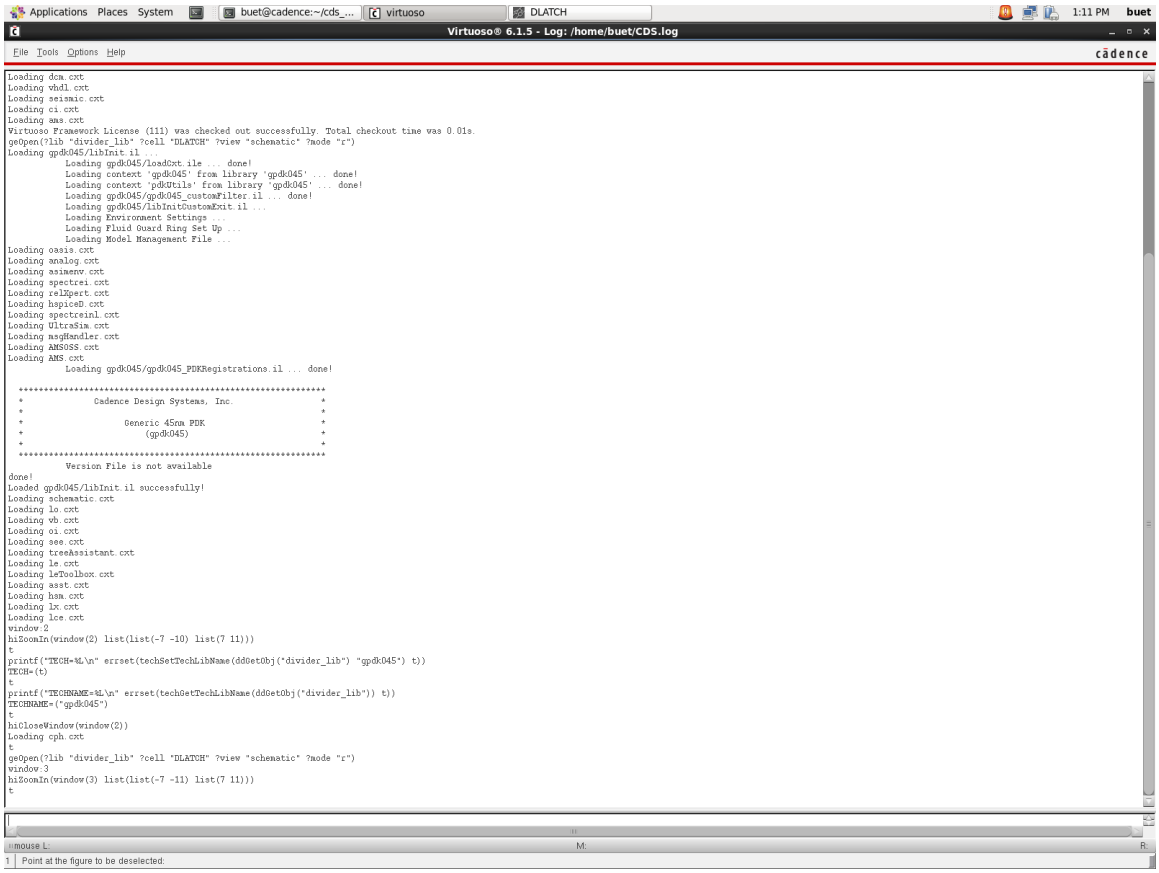


Figure 19: Virtuoso 6.1.5 session with the gpdk045 PDK loaded (CIW log, Cadence GUI screenshot).

NOT YET AVAILABLE: Figure 20 - additional figure 12 (file not found: screens/library_manager.png)

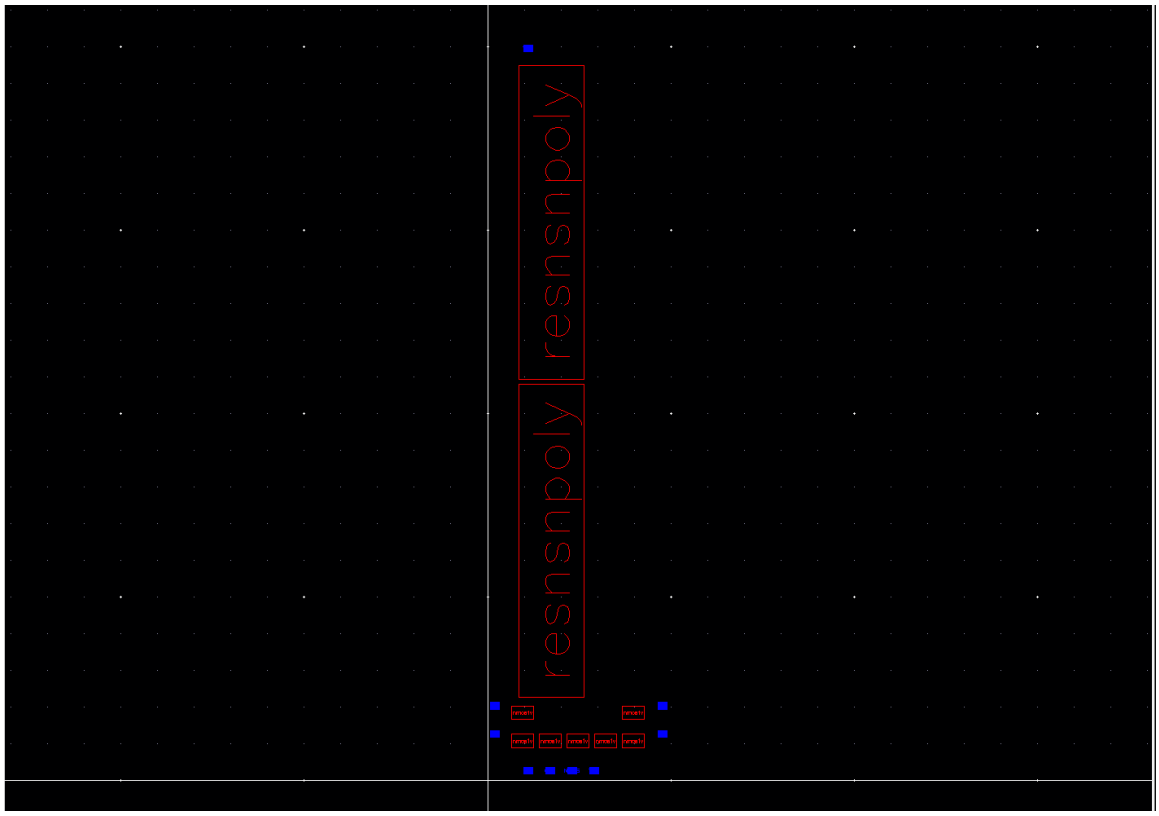


Figure 21: Virtuoso layout of the DLATCH: 7 nmos1v pcells, 2 resnsnpoly poly resistors and pin markers (Virtuoso Layout Suite screenshot, cropped to the layout canvas). Placement only; not routed.

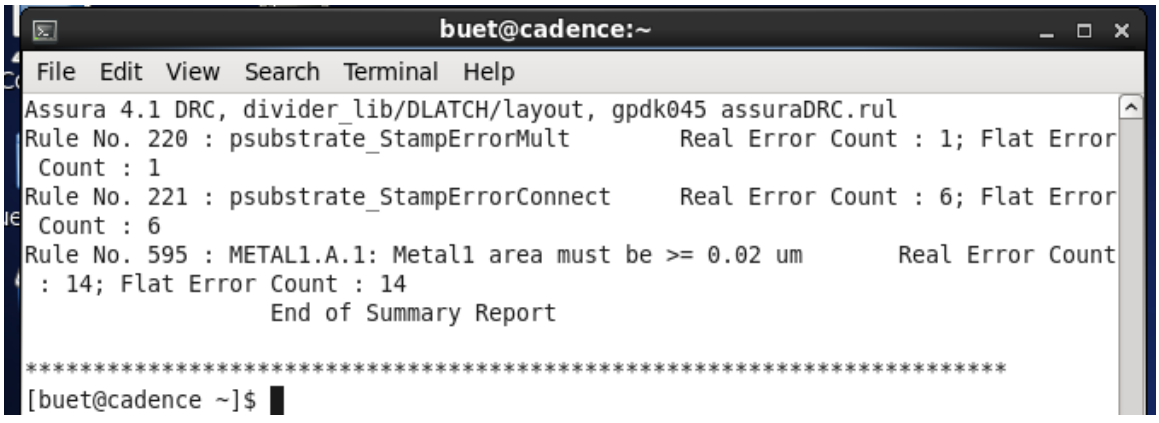


Figure 22: Assura 4.1 DRC tally for the DLATCH layout (divider_lib/DLATCH/layout, gpdk045 assuraDRC.rul): 21 errors in 3 rules (terminal capture).

CONCLUSION:

Both operating points work with the same topology and the stack rule. Design D at 1.4 GHz (m = 4, R = 3 kOhm) divides with 0.5413 mW, 25.99 ps delay, 14.067 fJ PDP and a 38.46 deg phase-step error, and passes 41 of 45 PVT points. Design B at 6 GHz (m = 4, R = 2 kOhm) divides with

0.4929 mW, 27.94 ps delay, 13.768 fJ PDP and a 3.78 deg phase-step error, and passes 21 of 45 PVT points.

The unsized circuit with R1 = Ron/2 does not divide at 1.4 GHz, which is consistent with the stack-resistance rule (its load is far below the threshold).

Limits. The tail M7 operates in triode, not saturation, so the paper's tail model is not reproduced. The phase-step error at 1.4 GHz (38.46 deg) is far above the paper's 3.75 deg, and only design B at 6 GHz comes close (3.78 deg). All 4 PVT failures of design D are at 125 C and 0.9 V. Design B, whose load lies inside the regeneration band, passes fewer PVT points (21 of 45 at 6 GHz and 27 of 45 at 1.4 GHz).

Layout status: placement only, and not clean or complete. The divider_core (2x2 DLATCH placement) and the DLATCH are placed but not routed, not extracted, not LVS-checked and not simulated after layout. The DLATCH Assura DRC gives 21 errors in 3 rules (7 substrate-stamp connectivity, 14 Metal1 minimum area). The post-layout waveform is not available: post-layout simulation not performed (layout not routed/extracted).

PROBLEMS FACED DURING THE IMPLEMENTATION:

Spectre aborted with ERROR (VACOMP-1008): Cannot compile ahdlcmi module library, and then a segmentation fault, for the gate-resistance Verilog-A style sub-model inside the gpdk045 device.

Cause: the Cadence VM is 32-bit CentOS 6.5, but the bundled perl (cdsPerl) and gcc are 64-bit and could not run, and no gcc was installed.

The guest disk had only about 100 MB free, so a sweep failed with No space left on device.

The unit-size circuit with R1 = Ron/2 does not divide at 1, 1.4 or 2 GHz.

The first division check counted false passes: outputs at 2 or 2.8 times the expected frequency with only 15 to 20 mV of swing.

The Spectre ASCII raw layout (first variable on the Variables line, several values per line) broke the first parser.

The 125 C corners failed for the marginal sizing.

The VM had no guest tools, so files were moved by virtual CD images and a virtual serial port with MD5 checks.

The phase error at 1.4 GHz is large (38.46 deg for design D), far from the paper's 3.75 deg.

Layout not completed. The layout is placement only: intra-latch routing, vias, clock trunks and substrate taps are not drawn. The DLATCH DRC gives 21 errors (7 substrate-stamp, 14 Metal1 minimum area), LVS and extraction were not run, and post-layout simulation was not performed.

PROBLEMS RESOLVED USING SOLUTIONS:

Installed gcc 4.4.7 and glibc-devel from the CentOS 6.5 vault RPMs, replaced the bundled perl wrapper by the system perl, and pointed the bundled gcc and ld symlinks at /usr/bin.

The measured values were stored as TSV and JSON files (nominal, sweep, PVT and analysis files), so each number in this report is read from one of those files.

The unsized design was replaced by the sizing concept: the load was set from the stack-resistance rule, giving design D (41 of 45 PVT points).

The division test was made strict: all eight outputs at f_in/2 within 0.1 percent, each swinging at least 0.1 V.

The parsers were fixed for the Spectre ASCII raw layout, and the PVT and sweep results were checked against their summary files.

Two operating points were chosen: 1.4 GHz, the paper's input frequency, and 6 GHz, the point with the smallest phase-step error. Each has its own PVT matrix.

Assura batch DRC now runs from a run file. The file needs avParameters with ?inputLayout ("df2" "divider_lib") ?cellName ?viewName ?rulesFile ?workingDirectory ?runName ?avrpt t, and is run as `assura file.rsf` (no avDRC call). This gave the DLATCH DRC tally in the Layout text.

ANY OTHER COMMENT:

The waveform and PVT figures are plots generated from Spectre data. They are labelled as plots, and no Cadence GUI screenshot is substituted for them.

The Ron value and the recorded division pairs are taken from LEDGER.md. The Ron value is not in a separate output file, so the report cites the ledger.

The two circuit drawings are generated from the gpdk045 Spectre netlist with the design D labels, so they agree with Tables 1 and 2. The earlier baseline drawings in report/figures are not used in this report.

The paper's constants are literature values, and the title-page register and name fields are left blank by design.

REFERENCES:

Y.-F. Kuo et al., "A Compact Multiphase Divide-by-Two CMOS Frequency Divider," IEEE Microw. Wireless Technol. Lett., vol. 35, no. 5, pp. 617-620, May 2025, doi: 10.1109/LMWT.2025.3528863.

Cadence Design Systems, Cadence GPDK045 v6.0 documentation (45 nm generic process design kit).

Cadence Design Systems, Spectre Circuit Simulator Reference, MMSIM 12.1.